Staff answers · shared work

Starter

Probability applies to each independent fertilisation; it does not force a fixed sequence in a small sample.

Worked example

A model gives a 50% chance of an outcome. Find the expected count in 30 trials.

Write 50% as 0.5.

Expected count = 0.5 × 30 = 15.

The model predicts 15. A real count can be more or less than 15.

Each trial has its own chance. Its result does not fix the next one.

Shared evidence task

Expected aa counts are 5, 5 and 20. Observed aa percentages are 35%, 15% and 22.5%. Variation is compatible with random sampling; these small datasets alone do not establish a different inheritance model.

Card placements

Card	Group
Model: 5 aa in 20	Model prediction
Model: dominant trait in 75%	Model prediction
Sample: 7 aa counted	Sample count
Next sample: 3 aa counted	Sample count
Four offspring must give one aa	False claim
The next one must make the counts fit	False claim

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Staff answers · Supported

Arrival

1. Fraction: 25%.

2. Genotypes: 1 AA : 2 Aa : 1 aa.

3. Phenotypes: 3 dominant : 1 recessive.

4. Independence: No.

Independent / application

1. Calculate: 5, 5 and 20.

2. Convert: 35% and 22.5%.

3. Explain: Random variation can produce a count above or below the expectation; more suitable evidence is needed to judge the model.

Two-check exit

1. Calculate: 1/4 × 80 = 20; this is an expectation, not a guarantee.

2. Interpret: Random variation means observed counts need not equal the expected proportion exactly.

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Staff answers · Independent

Arrival

1. Fraction: 25%.

2. Genotypes: 1 AA : 2 Aa : 1 aa.

3. Phenotypes: 3 dominant : 1 recessive.

4. Independence: No.

Independent / application

1. Calculate: 5, 5 and 20.

2. Convert: 35% and 22.5%.

3. Explain: Random variation can produce a count above or below the expectation; more suitable evidence is needed to judge the model.

Two-check exit

1. Calculate: 1/4 × 80 = 20; this is an expectation, not a guarantee.

2. Interpret: Random variation means observed counts need not equal the expected proportion exactly.

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Staff answers · Aspire

Arrival

1. Fraction: 25%.

2. Genotypes: 1 AA : 2 Aa : 1 aa.

3. Phenotypes: 3 dominant : 1 recessive.

4. Independence: No. Accept a scientifically correct reason that uses the relevant model or evidence.

Independent / application

1. Calculate: 5, 5 and 20.

2. Convert: 35% and 22.5%.

3. Keep events independent: 25%. Each fertilisation is independent, so earlier outcomes do not use up or force a later genotype.

Two-check exit

1. Calculate: 1/4 × 80 = 20; this is an expectation, not a guarantee.

2. Keep events independent: 25%. Each fertilisation is independent, so earlier outcomes do not use up or force a later genotype.

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Staff · source and teaching care

Weekly binding: Complete a genetic cross (digital Punnett square).

Exact pupil goal: Convert genetic probabilities between fractions, ratios and percentages, and explain why observations vary.

Source: Launch/W8-W13_2026-27/SCI_L_W13L3_Inheritance_Probability_Do.html; revision c4ca8ffab98e6186a9ef2fed3921644b177f0d6d.

Workbook: _authoring/science_2026-27/_toolchain/scheme_references/LAUNCH_SOW.xlsx; LAUNCH Weekly - Autumn B43 Aut2·W5; C43 Complete a genetic cross (digital Punnett square).

Sources are inherited from the verified lesson. No new specification alignment is asserted. [AWARD] [QUAL] names remain masked. No biological or personal family data is used.

Hearing both pupils matters. A correct click is not evidence of reasoning. Re-model the uncertain step for one pupil while the other attempts a fresh example. Preserve the original answer and the prompt used.

A genetic counsellor explains chance without promising a fixed outcome. This is a career link, not advice about a person.

Model care: This is a simple model for one gene. A dominant allele is not stronger, better or more common. Each fertilisation is a new chance. Do not use real family traits or ask for personal or medical details.

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